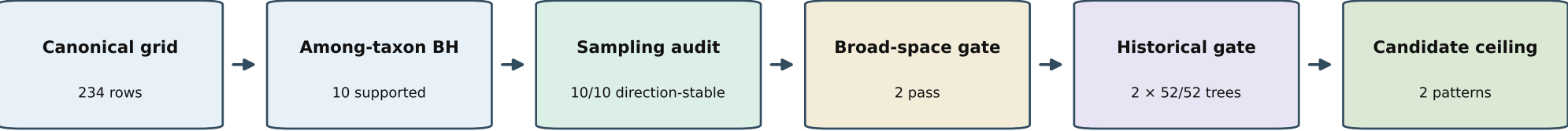
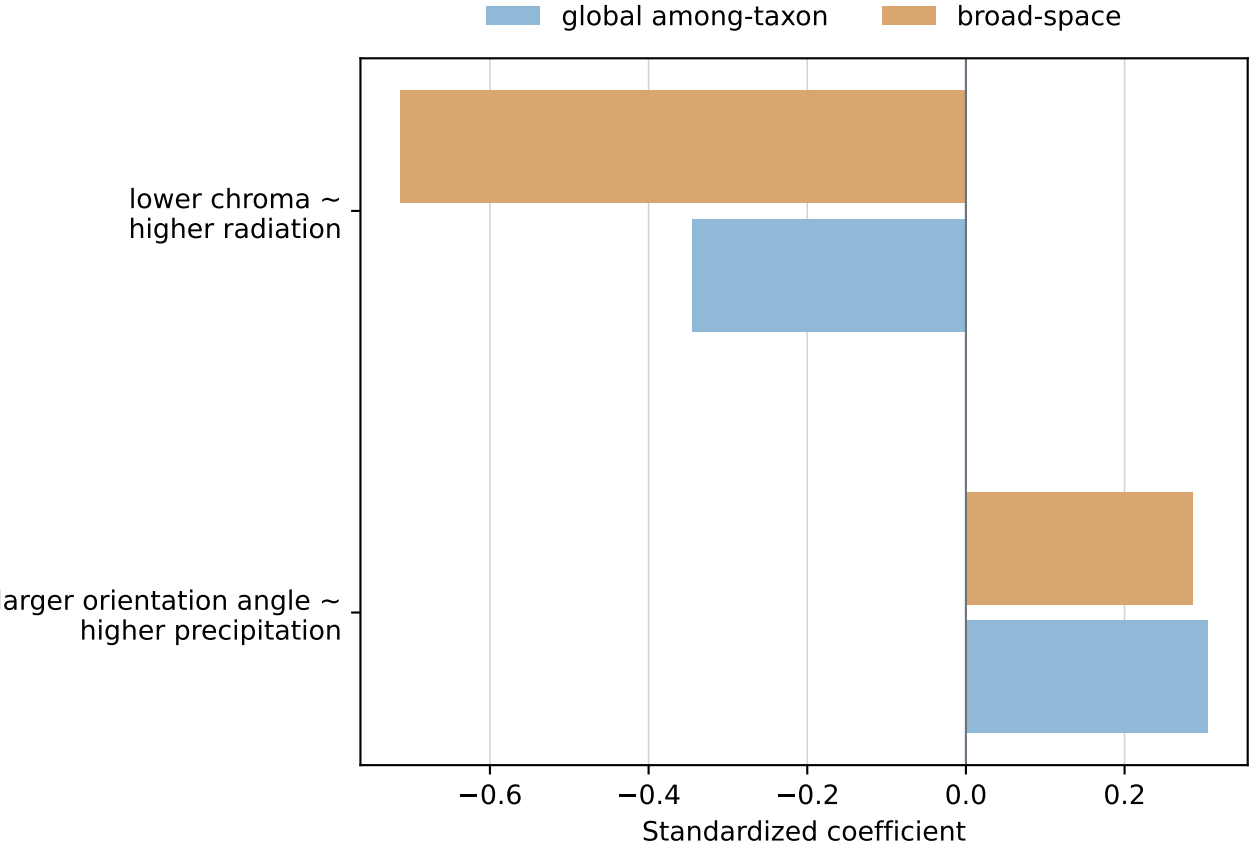


(a) Sequential attrition: why only two among-taxon rows remain



Sampling stability annotates selected rows; promotion requires broad-space control followed by historical-placement sensitivity.

(b) Direction after broad-space control



(c) Gate evidence retained in the frozen ledger

**lower chroma ~ higher radiation**

global  $\beta = -0.345$ ;  $q = 0.006$

sampling minimum ratio = 0.615; broad-space  $\beta = -0.712$

spatial  $P = 0.001$ ; residual Moran  $P = 0.217$ ; trees = 52/52

**larger orientation angle ~ higher precipitation**

global  $\beta = +0.304$ ;  $q = 0.021$

sampling minimum ratio = 0.781; broad-space  $\beta = +0.286$

spatial  $P = 0.017$ ; residual Moran  $P = 0.082$ ; trees = 52/52